Maharaja Surajmal Institute of Technology

MINOR PROJECT

Melanoma Detection System

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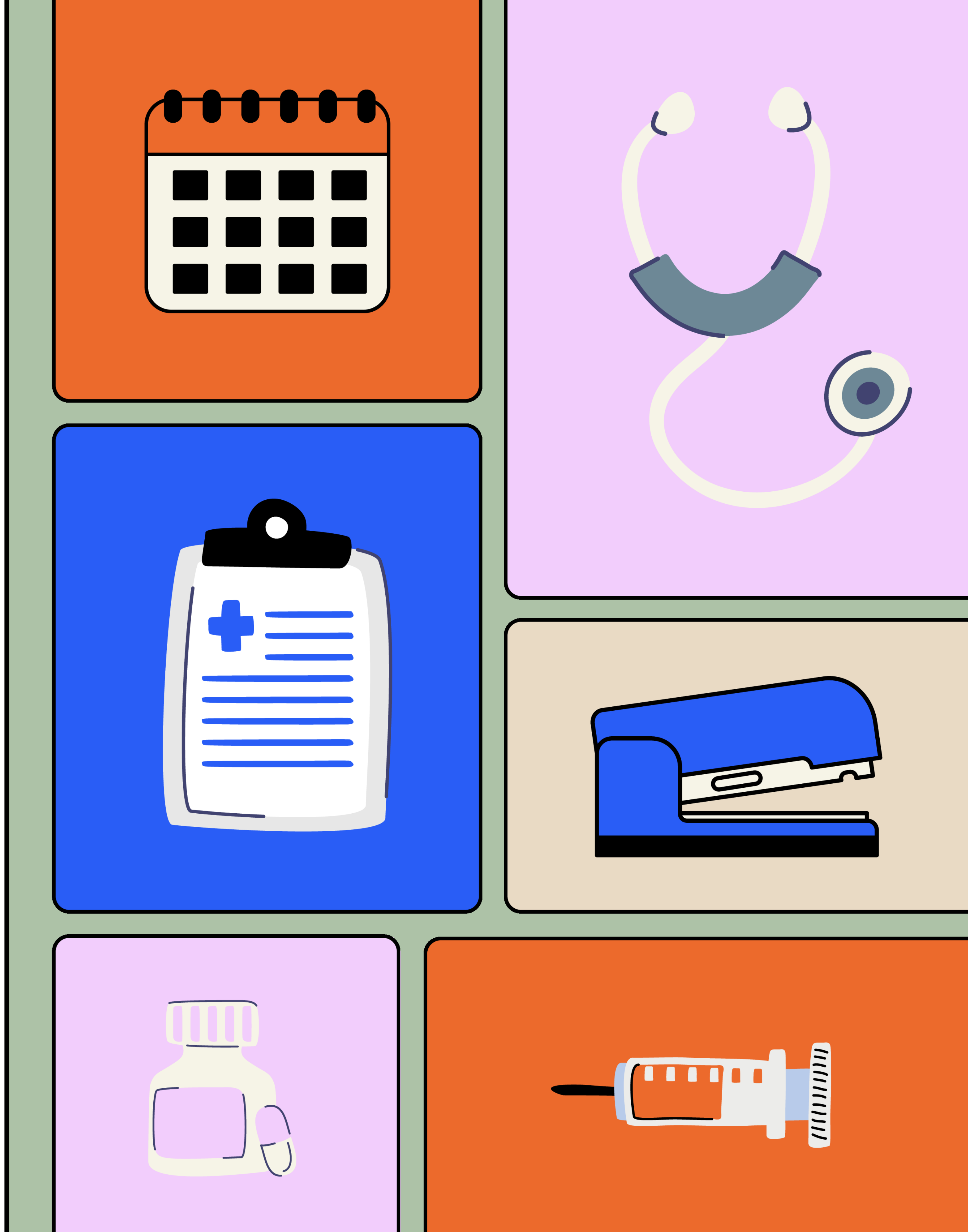

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OVERVIEW

- Main objective: To detect presence of melanoma using high resolution skin lesion images.
- Developed a deep learning model with the help of CNN.
- Utilized a publicly available dataset (ISIC) for training and testing.
- Utilizing various tools like Keras, TensorFlow, scikit, classifier etc to aid the process.

Melanoma

- A type of skin cancer caused in the pigment producing skin cells known as ‘melanocytes’.

Objectives Achieved

- Dataset Collection: Melanoma and non-melanoma images gathered.
- Data Pre-processing: Resizing, normalizing and augmenting data.
- Training two CNN models with accuracy 96.52% and 98.16% respectively.
- Model ensembling with final accuracy of 98.52%

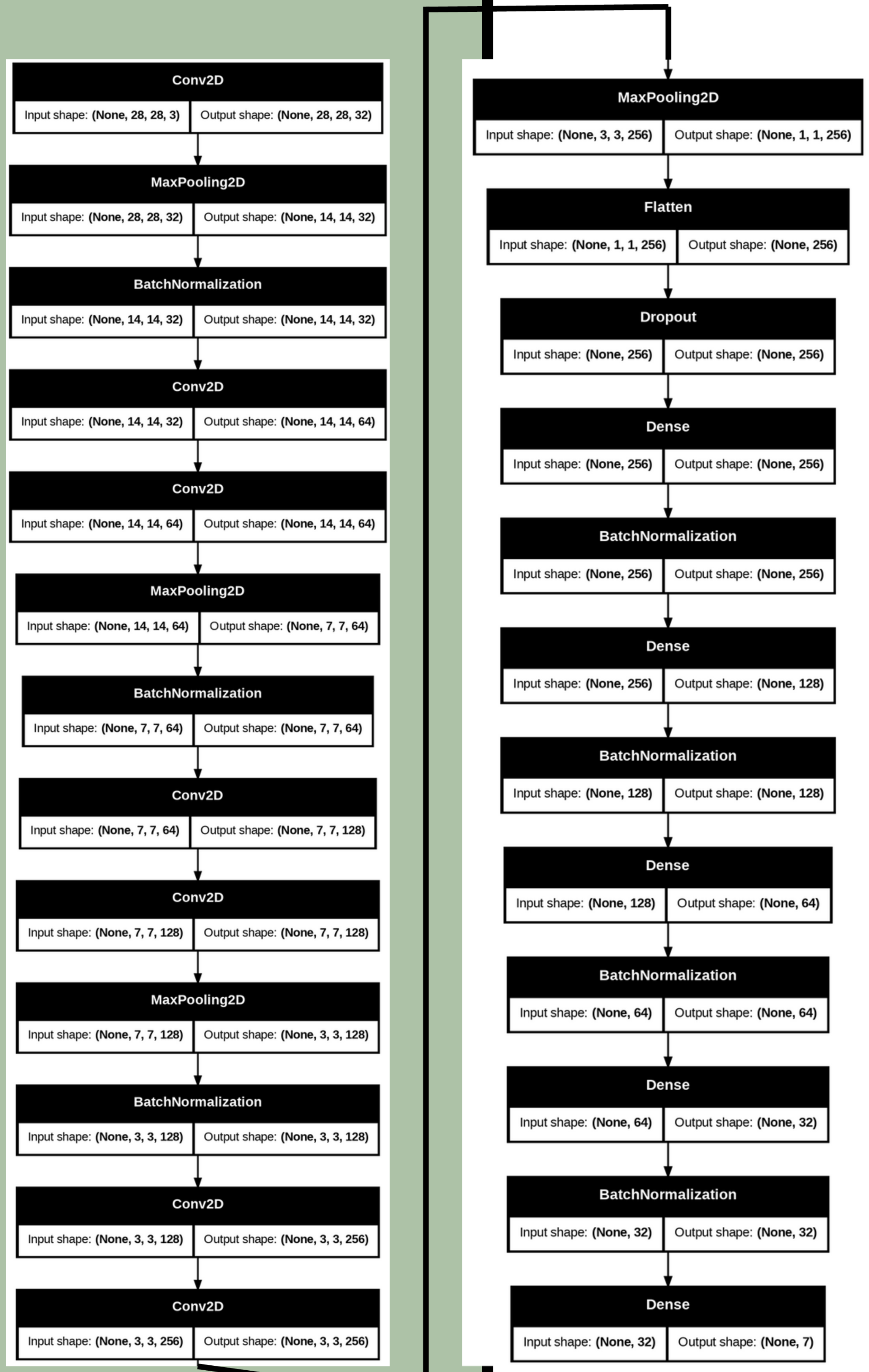


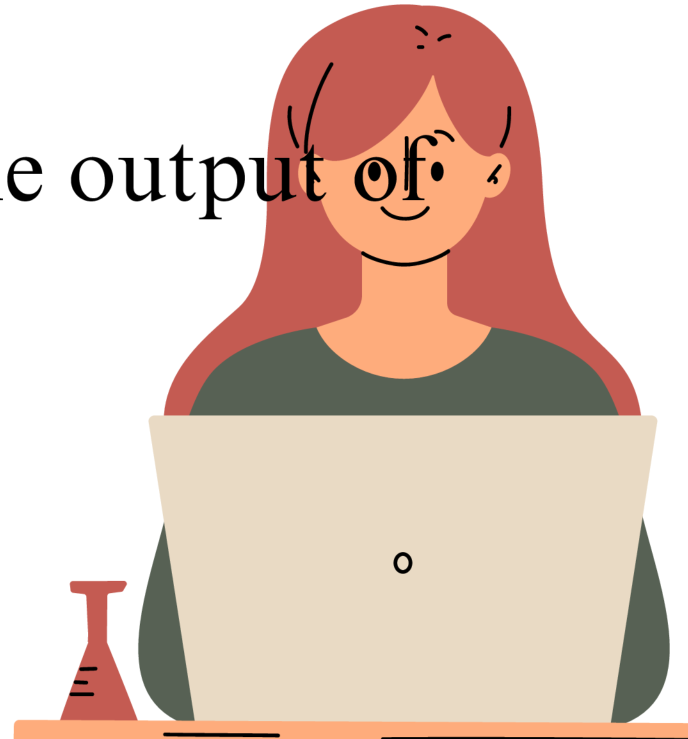
Fig. 1: Model A summary

CNN Layers

CNN is an abbreviation of Convolutional Neural Networks.

It is a multilayered network and its basic layers are:

- **Conv2D**- feature extraction.
- **MaxPooling2D**- reduces the spatial dimensions of the feature maps.
- **Batch Normalization**- normalize the output of the previous layers.



CNN Layers

- **Flatten layer**- flattens 2D feature maps into a 1D vector.
- **Dropout Layer**- Removes a portion of neurons.
- **Dense Layer**- Predictions are made.
- **Output Layer**- Output

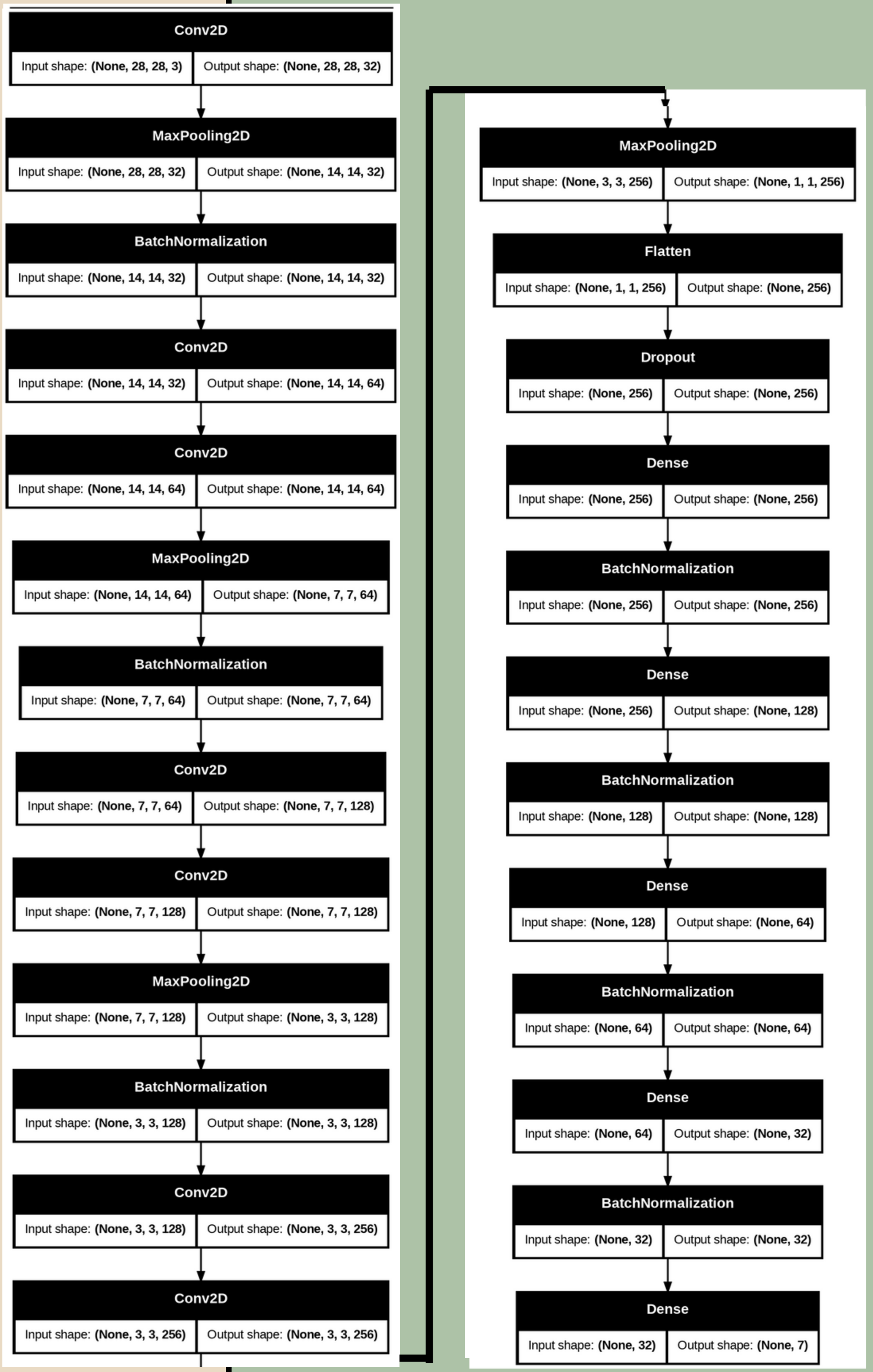


Fig. 2: Model B summary

Models

- Two CNN models Model A and Model B are created with similar architecture differing in:

Table 1: Model and Model B functions

	Model A	**Model B**
Loss Function	Categorical Cross Entropy	Mean Squared Error
Activation Function	ReLU	Leaky ReLU
Weight Initializer	Normal	Truncated Normal

Ensemble Model

Both Model A and Model B are ensembled together and converted to a TFLite format.

Table 2: Training and testing accuracies

S.no.	Models	Training Accuracy	Testing accuracy
1	Model A	97.8	96.52
2	Model B	99.08	98.16
3	Ensemble model	99.5	98.5

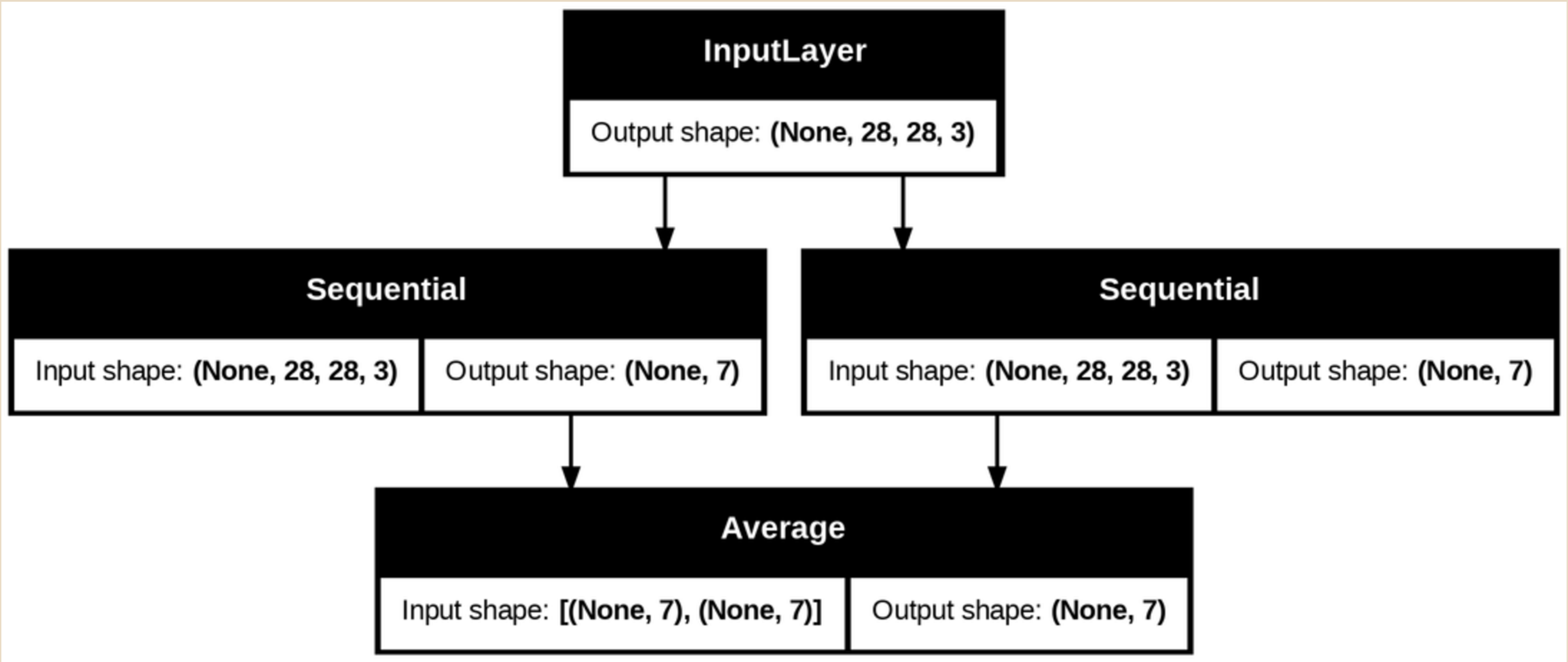


Fig. 3: Ensemble Model Architecture

Table 2 shows the testing and training accuracies of model A, model B and the ensemble model.

Through this, we can say the aggregated model had the best performance.

Results

The ensemble model received commendable results with the accuracy of 98.52%.

- Confusion matrix, fig. 4, compares the actual target values with the predicted values.

- Class 6 i.e melanoma: 1359 correct predictions with 31 inaccurately predicted as class 4.

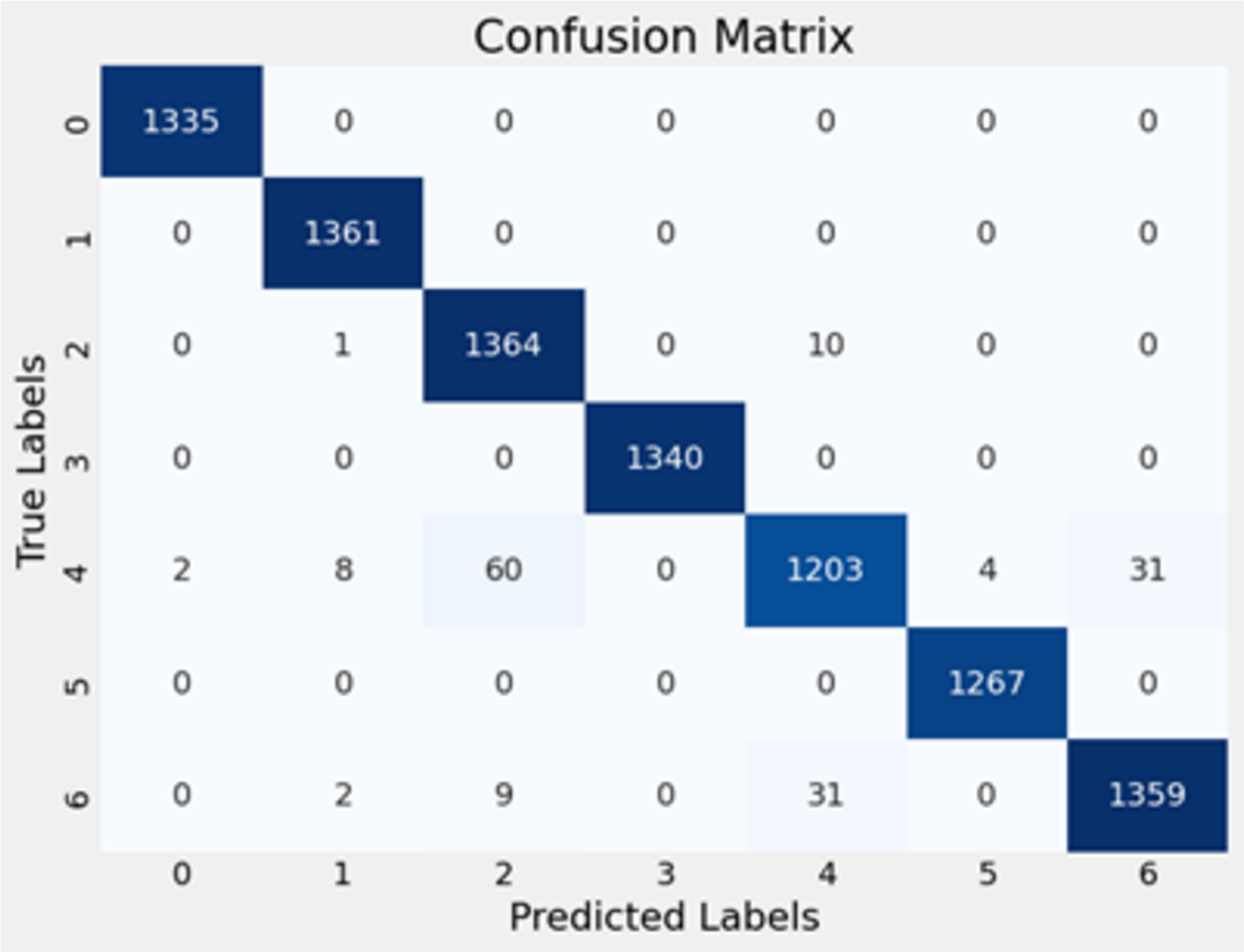


Fig. 4: Confusion matrix of ensemble model

Conclusion

In conclusion, Our research project successfully developed an ensemble model that achieved an impressive accuracy of 98.52%, surpassing the individual performances of Model A (96.52%) and Model B (98.16%).

By utilizing an averaging ensemble technique, we effectively combined the strengths of both models, leading to improved accuracy and better generalization on unseen data. This demonstrates the potential of ensemble learning to enhance predictive performance in real-world applications. Future work could explore advanced ensembling methods like stacking and boosting to further refine accuracy and model robustness.

```
Accuracy: 0.9852
Precision: 0.9852
Recall: 0.9852
F1-Score: 0.9852
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Thank you !